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Lesson Notes

MODULE 4 | LESSON 4

EXTREME INNOVATIONS IN VALUE AT RISK MODELING

Reading Time	25 minutes
Prior Knowledge	Leverage, Correlation, GARCH
Keywords	EGARCH, GJR-GARCH, AP-ARCH

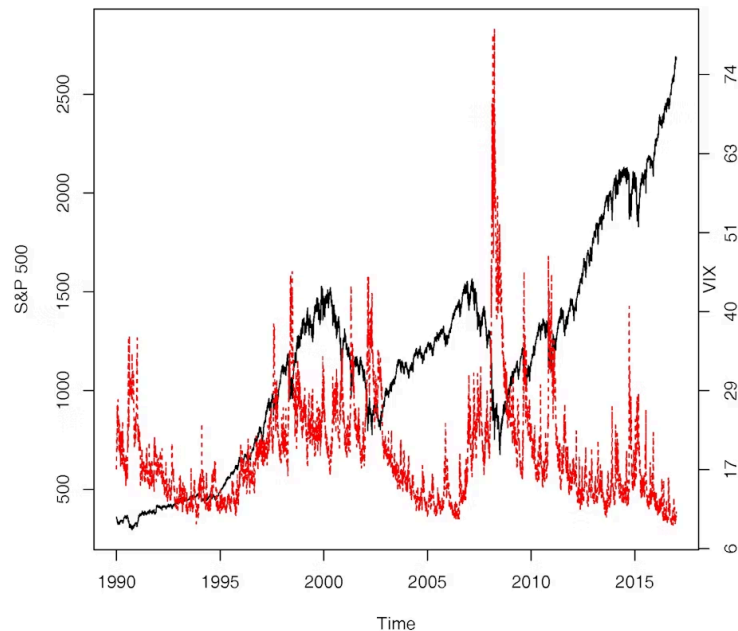
In the previous lesson, we used EVT in combination with the standard GARCH model, but in this lesson, we use the more recently developed versions of GARCH to arrive at even better estimates.

Throughout this and the last module, we have referenced a wide variety of GARCH models without distinguishing between them in meaningful ways or discussing them in much detail. As we can see now, though, these extensions and variations of GARCH make their distinct contributions to the forecasting of volatility required by Value at Risk estimation.

1. EGARCH

Introduced in 1991, Exponential GARCH (EGARCH) stands out from other models in that it captures what Omari refers to as a “leverage effect” (852), but we can conceive of it more generally as the asymmetric effect of the sign of previous returns shocks. (Note: Omari et al. refer to the return shock as the “innovation term” (851). There is in fact a negative correlation between the recent shock and future volatility, such that a negative shock makes a larger contribution to future volatility than a positive shock. It was believed that the increased future volatility was driven by the increase in leverage that mathematically followed from a negative shock.

Figure 1: Leverage Effect, or the Negative Correlation between Returns and Volatility (01/02/1990–12/29/2017; black: S&P 500 Index; red: VIX)



Source: Sun, Yiguo, and Ximing Wu. "Leverage and Volatility Feedback Effects and Conditional Dependence Index: A Nonparametric Study." *Journal of Risk and Financial Management*, vol. 11. no. 2, 2018, <https://www.mdpi.com/1911-8074/11/2/29/htm>.

2. "Leverage Effects"

Recall from Financial Markets our discussions of leverage and take for example the equity multiplier measure of leverage, simply assets divided by equity. Assume we start with 100 rupiah in assets and 10 rupiah in equity, meaning leverage is 10x. If our returns shock takes the form of an overall asset price decline of merely 1%, our leverage increases 10% to 11x because the equity multiplier is now 99 rupiah divided by 9 rupiah. Now place this example in the context of our many discussions of the "fire sale" phenomena: If we are mandated to keep our equity multiplier at or below 10x, then we need to sell assets in order to get back in compliance with that mandate, putting additional downward pressure on prices. The scenario where this vicious cycle continues was therefore thought to explain the asymmetric effect of "good news" vs. "bad news" in the form of returns shocks. However, Vlab notes that the idea that the asymmetry actually being an effect of leverage alone has been disproved and there must be more to this story (Engle). Regardless of the full explanation, EGARCH accounts for this asymmetry.

The equation used to fit the parameters of the EGARCH model involves the log variance, instead of simply the variance. Both sides of the equation must effectively be exponentiated, giving Exponential GARCH its name.

An interesting result of the lack of parameter restrictions for this model in comparison to the positivity and non-negativity restrictions in the standard model: The γ parameter used to model the asymmetry is itself generally negative, but the variance is nonetheless necessarily positive given the domain of the logarithmic function used to estimate the entire parameter set.

3. GJR-GARCH

Glosten-Jagannathan-Runkle GARCH (GJR-GARCH) also accounts for the asymmetry we have been discussing, but by incorporating an indicator variable for whether the shock is negative:

$$\sigma_t^2 = \omega + (\alpha + \gamma I_{t-1})\epsilon_{t-1}^2 + \beta\sigma_k^2$$

I_{t-1} is zero if the most recent return is greater than or equal to the mean return; I_{t-1} is one if the most recent return is less than the mean return—which of course means the shock term has an increasing effect on the forecasted variance. Given the (non-logarithmic) form of the main GJR-GARCH equation, the parameter restrictions of the standard GARCH model apply; even the γ parameter, which incorporates the “bad news” effect is positive.

Also interesting is the fact that volatility is mean-reverting so long as the following holds:

$$\alpha + \gamma/2 + \beta < 1$$

And more specifically, the unconditional variance under such circumstance can be defined as follows:

$$\sigma^2 := \text{Var}(rt) = \omega / (1 - \alpha - \beta - \gamma/2)$$

V-Lab makes this point and explains that the $\frac{1}{2}$ multiplier is the result of the “the assumption that the conditional distribution of the returns is symmetric around μ ” since they are assumed to be normal.

4. APARCH

We include the symmetric Power ARCH (APARCH) model in this brief discussion, though it is not studied and tested by Omari et al., because it is special in several ways.

$$\sigma_t^\delta = \omega + \sum_{i=1}^p \alpha_i (|\varepsilon_{t-i}| - \gamma_i \varepsilon_{t-i})^\delta + \sum_{j=1}^q \beta_j \sigma_{t-j}^\delta$$

1. It can be viewed as a generalization of other important (G)ARCH models, as can be seen in these examples indicated by (there are others as well):
 1. for $\delta = 2, \gamma = 0, \beta = 0$ you have the ARCH(1) model (V-Lab)
 2. for $\delta = 2, \gamma = 0$: you have the GARCH(1,1) model (V-Lab)
 3. for $\delta = 2$: you have the GJR-GARCH(1,1) model (V-Lab)
 4. for $\delta = 0$: you have the EGARCH(1,1) model (Schmidt 11)
2. Given this versatility, it will come as no surprise that APARCH also models the asymmetry we have discussed.
3. Since the power term δ is just another parameter to estimate in the model-fitting process and the only restriction is that it be a positive real number, its inclusion should improve the goodness of fit by virtue of the flexibility it provides the model (Schmidt 11): While the power terms for GARCH and GJR-GARCH are “stuck” at two, and for EGARCH at zero, the APARCH power term has been found to be 1.43 and 1.524 for U.S. stock returns according to different studies (McKenzie and Mitchell 3).

5. Conclusion

We discussed extensions of the GARCH model and combining these with EVT.

References

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